

Data Transformation Techniques in Machine Learning

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What is Data Transformation?

- Data transformation is the process of converting data from its original format into a format suitable for analysis.
- It involves modifying the structure, format, or values of data to enhance its quality and usability.
- Typically follows the data cleaning phase.
- Example: Changing string labels to numerical values, scaling numeric features.

Pipeline: Raw Data → Cleaning → **Transformation** → Modeling

Why Do We Need Data Transformation?

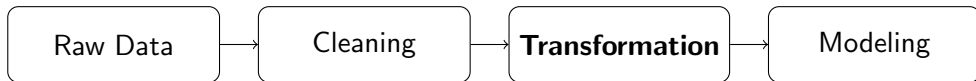
- ML algorithms often expect numeric and standardized inputs.
- Raw data might contain inconsistencies, missing values, or varying formats.
- Helps in feature comparability by placing them on the same scale.
- Reduces computational complexity and improves convergence speed.
- Improves accuracy and efficiency of ML models.

Techniques Covered in This Lecture

In this lecture, we will cover the following common data transformation techniques:

- **Mapping** – Converting categorical variables to numeric format.
- **Quantization** – Grouping continuous variables into discrete bins.
- **Binarization** – Transforming values into binary format (0 or 1).
- **Normalization** – Rescaling values to a fixed range, typically $[0, 1]$.
- **Scaling (Standardization)** – Adjusting data to have zero mean and unit variance.

Where Transformation Fits in the Pipeline



- Converts categorical values to numerical representations.
- Often called label encoding.
- **Example:**
 - Gender: Male $\rightarrow$ 1, Female $\rightarrow$ 0
 - Education Level: High School $\rightarrow$ 1, Bachelor $\rightarrow$ 2, Master $\rightarrow$ 3
- Essential for algorithms that require numeric input.

- Converts continuous data into discrete categories.
- Useful for reducing noise and model complexity.
- **Example: Age Quantization**
 - 0–12 → Child
 - 13–19 → Teen
 - 20–59 → Adult
 - 60+ → Senior
- Often used in decision trees and rule-based models.

- Transforms numerical or categorical values into binary (0/1).
- **Example: Income Level**
 - If income $> 50k \rightarrow 1$
 - Else $\rightarrow 0$
- Useful in classification tasks and for simplifying features.

Normalization

- **Definition:** Normalization is the process of rescaling feature values so they fall within a specific range, typically $[0, 1]$.
- **Purpose:**
 - Makes features equally weighted in distance-based models (like KNN or K-Means).
 - Prevents features with larger ranges from dominating the learning process.
- **Common Method: Min-Max Normalization**

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Where:

- X = original value
- X_{min}, X_{max} = minimum and maximum values in the column
- **Example:** Suppose heights range from 150 cm to 200 cm. If someone is 170 cm:

$$\frac{170 - 150}{200 - 150} = 0.4$$

- **When to Use:**

- When you need values bounded in $[0, 1]$

Scaling (Standardization)

- Centers features around mean = 0 and std = 1.
- Useful for models sensitive to feature scales (e.g., SVM, logistic regression).
- Formula:**

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

- Example:** A value of 20 with mean = 10, std = 5 becomes:

$$\frac{20 - 10}{5} = 2.0$$

1. Min-Max Scaling:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

- Best when you want a bounded range $[0,1]$.
- Preserves relationships but not outliers.
- Use when the distribution is not Gaussian.

2. Standard Scaling (Z-score)

$$X_{scaled} = \frac{X - \mu}{\sigma}$$

- Best when data is normally distributed.
- Makes comparison between different features fair.
- Suitable for distance-based models.

3. Robust Scaling:

$$X_{robust} = \frac{X - Q_2}{Q_3 - Q_1}$$

- Based on median (Q_2) and IQR ($Q_3 - Q_1$).
- Less sensitive to outliers.
- Use when data contains extreme values or is not normally distributed.

Summary Table

Technique	Input	Output	Use Case
Mapping	Categorical	Numeric	Encode labels
Quantization	Continuous	Discrete bins	Simplify ranges
Binarization	Numeric/Cat	Binary	Threshold-based flags
Normalization	Continuous	[0,1] range	Compare feature values
Scaling	Continuous	Mean=0, Std=1	For normal distributions
Robust Scaling	Continuous	Median-centered	Outlier resistance

Practice Problems (1/3) – Conceptual

- Q1: Why is it important to normalize or scale data before using algorithms like KNN or SVM?
- Q2: Give one scenario where quantization might reduce model performance.
- Q3: What could be a problem with mapping categorical features that are nominal (like city names) into numeric values?
- Q4: How can scaling improve model convergence?
- Q5: Why is robust scaling preferred in datasets with outliers?
- Q6: Can normalization and standardization be used together? If so, when?

Practice Problems (2/3) – Identify the Technique

- Q7: You have a column with age values. You want to convert it into child/teen/adult/senior. What technique will you use?
- Q8: You are working with a binary classification model. Income above 40k is classified as high income. What transformation suits this?
- Q9: You have a column with height in cm. You want all values between 0 and 1. What should you apply?
- Q10: You want to convert 'education level' from strings to integers. Which transformation applies?
- Q11: When should you prefer robust scaling over standard scaling?
- Q12: What transformation would you apply to make sure no feature dominates in a clustering model?

Practice Problems (3/3) – Coding Challenge

- Q13: Given a dataset with the column 'education' = ['High School', 'Bachelor', 'Master'], write Python code to map it to [1, 2, 3].
- Q14: Normalize a list of values [10, 20, 30, 40] using min-max normalization.
- Q15: Use sklearn to standardize the list [5, 10, 15].
- Q16: Apply robust scaling to the list [1, 100, 3, 2, 5, 1000].
- Q17: Use pandas to binarize a 'score' column with a threshold of 50.
- Q18: Write a function to quantize temperature into 'Cold', 'Warm', 'Hot'.